



Amazon tools and services in North Star

27 tools and services across 6 areas | Agentic AI, IT Expert System | David Dunmeyer and Biswa

AI AND AGENTS	
Amazon Bedrock	Hosts and runs the models; every agent call goes through it
Amazon Nova 2 Lite	The model behind the coordinator and all three specialists
Amazon Bedrock Knowledge Bases	Retrieves policy text so answers cite a document and section
Amazon Titan Text Embeddings V2	Turns the policy documents into vectors for search
Strands Agents	Open-source agent framework from AWS; the agents and their tools are built with it
DATA AND STORAGE	
Amazon DynamoDB	One table for employee records and confirmed requests
Amazon S3	Stores the six policy documents
Amazon S3 Vectors	Stores the vectors the Knowledge Base searches
HOSTING	
Amazon EC2	The t3.small server that runs the app
Amazon Linux 2023	The server's operating system image
Amazon EBS	The server's 12 GB disk
Elastic IP	A fixed public address, so the QR code survives restarts
Amazon VPC security group	Opens port 80 only; no SSH
OPERATIONS, REUSED FROM OUR EC2 SCHEDULER PROJECT	
Amazon EventBridge Scheduler	Triggers the 8:00 AM start and 6:00 PM stop
AWS Lambda	The functions that start and stop the server
Amazon SNS	Emails us when the server starts or stops
Amazon SQS	Dead-letter queue that catches a failed schedule run
Amazon CloudWatch Logs	Logs from the scheduler's functions
AWS CloudFormation and AWS SAM	Define the scheduler as code
AWS Budgets	A \$5 monthly budget with alerts
SECURITY AND ACCESS	
AWS IAM	Least-privilege roles for the server and the Knowledge Base
AWS STS	Short-lived credentials; no keys stored in code or on the server
AWS Systems Manager	Looks up the current server image; Session Manager gives shell access without SSH
DEVELOPER TOOLS	
AWS CLI v2	Sign-in with aws login, and setup from the terminal on Mac and Windows
AWS SDK for Python (Boto3)	How the app and scripts call AWS
AWS Management Console	Building and inspecting resources by hand
AWS CloudShell	Browser terminal used for early checks